

DIR Assignment

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1. Horner's method 11001101 to decimal

1		$x2 =$	2
+ 1	= 3	$x2 =$	6
+ 0	= 6	$x2 =$	12
+ 0	= 12	$x2 =$	24
+ 1	= 25	$x2 =$	50
+ 1	= 51	$x2 =$	102
+ 0	= 102	$x2 =$	204
+ 1	= 205	$x2 =$	205

2. Horner's method BAD1 → decimal

B = 11

A = 10

D = 13

1 = 1

11		$x16 =$	176
+ 10	= 186	$x16 =$	2976
+ 13	= 2989	$x16 =$	47824
+ 1	= 47825		47825

3. repeated division 244 → binary

244	÷ 2	123	0
123	÷ 2	61	1
61	÷ 2	30	1
30	÷ 2	15	0
15	÷ 2	7	1
7	÷ 2	3	1
3	÷ 2	1	1
1	÷ 2	0	1

$$244_2 = 1111\ 0110$$

4. repeated division 50000 $\rightarrow$ hexadecimal

50000	$\div 16$	= 3125	0	0
3125	$\div 16$	= 195	5	5
195	$\div 16$	= 12	3	3
12	$\div 16$	= 0	12	C

$$50000_{10} = C350$$

5. C350 $\rightarrow$ binary

C 1100
3 0011
5 0101
0 0000
1100
8421

$$C350 = 1100\ 0011\ 0101\ 0000$$

6. 2's complement -123 $\rightarrow$ binary

$$128 - 123 = 5$$

1111 1111 1000 0101

7. $\uparrow$ in hex

1111 1111 1000 0101
F F 8 5

$$FF85$$

8.a. max. # of distinct VINs

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excluding (1, 0, 0) - cases

$$36 - 3 = 33$$

26 letters

10 digits

36 possibilities

excluding (1, 0, 0, 0, 2, 0) - cases

$$WMI = (33)^3$$

$$VD = (33)^5$$

$$CD = 0-9, X = 11$$

$$MY = 21 + 9 = 30$$

$$PC = 33$$

$$SN = (33)^6$$

$$36 - 6 = 30$$

1-17 digits

$$(33)^3 \times (33)^5 \times 11 \times 30 \times 33 \times 33^6$$

$$= (33)^{14} \times 11 \times 30 \times 33$$

8.b. min. # of bits req. for above

$$\log_2((33)^{14} \times 11 \times 30 \times 33) = 84.032234$$

85 bits min.

$$\frac{WMI}{33 \times 33 \times 33} \times \frac{VD}{33 \times 33 \times 33 \times 33 \times 33} \times \frac{CD}{33} \times \frac{MY}{11} \times \frac{PC}{30} \times \frac{SN}{33 \times 33 \times 33 \times 33 \times 33} = \frac{33^8 \times 11 \times 30 \times 33^6}{33^8 \times 11 \times 30 \times 33^6}$$

$$33^8 \times 11 \times 30 \times 33^6$$